46-923, Fall 2026: Homework #1

Due October 29 at 11:59 PM

Access OptionMetrics on WRDS. At the top level of OptionMetrics, you will see a link named "Options." Click on this, and then choose "Option Prices" in the list on the following page. Construct a .csv file consisting of all of the options information from 2024 for AAPL. Select all of the variables listed under "Option Price Information," **except for** "forward_price", "root", and "suffix". WRDS inludes three additional variables by default, including "issuer", "exercise_style", and "index_flag". There should be 26 variables total.

Create a Python notebook which takes the following steps.

1. Read in the data set. How many rows are there?
2. Make a list of all of the variables, and provide a description of each. In some cases the description provided on WRDS will be sufficient, but in others they are unclear. The resulting descriptions should provide enough detail to make it clear what is in the data set, aimed at someone who has a reasonable understanding of options.
3. Correctly format each of the date columns (there are multiple in this data set). The date of the price should be chosen as the row indices.
4. Transform any columns which are categorical and/or ordinal.
5. Propose and execute two different tests of conditions you believe that the data set should satisfy.
6. Which variables have at least one missing value? Can you determine the source of that missingness?
7. Focus only on the option with ID 156807233. Create a plot showing how the price of this option changes over the course of the year. Use the midpoint of the best bid and offer prices.
8. Again using option 156807233, create a scatter plot that compares the "moneyness" of the option with its trading volume on a daily basis. Calculate the moneyness as the ratio of the price of stock to the strike price. (You will have to obtain the closing price for the stock from another source.)